DRIVILX

Comprehensive Full-Stack System Test Report

On-Demand Professional Chauffeur & Driver Booking Ecosystem

Academic Milestone: SEM 9 Capstone Project	**Target Environment:** Live Localhost & MongoDB
Evaluation Date: October 22, 2026	**Overall Status:** **100% OPERATIONAL (PASS)**
Backend Stack: Django 5.2.5 REST + SimpleJWT	**Frontend Stack:** React 19 + Vite + TailwindCSS

19 / 19	100%	3 Portals	0 Failures
FEATURES TESTED	PASS RATE	USER | DRIVER | ADMIN	ZERO CRITICAL DEFECTS

1. Executive Summary & Test Objectives

DRIVILX is an enterprise-grade chauffeur booking platform engineered to solve urban mobility needs through specialized on-demand driver hire services (Hourly Chauffeur, City Transfer, Outstation Round-Trip, and Emergency Reliever). The system delivers three tailored stakeholder experiences: **Customer Portal** for real-time ride reservations, fare estimation, and driver tracking; **Driver Partner Cockpit** for duty management, incoming request dispatch, and earnings tracking; and **Platform Administrator Console** for fleet supervision, KYC moderation, and financial audits.

This evaluation report validates end-to-end multi-role system functionality under live execution. Both core servers were tested simultaneously: the Django REST backend connected to a local MongoDB cluster (seeding automated test accounts on startup), and the React 19 Vite client delivering sub-second response times.

Component	Runtime / Tech Stack	Port / Address	Verified Status
Backend API	Django 5.2.5 + Django REST Framework + SimpleJWT	http://localhost:8000/api/	**LIVE / HEALTHY**
Frontend Client	React 19 + Vite 6 + TailwindCSS Responsive	http://localhost:5173/	**LIVE / HEALTHY**
Database Engine	MongoDB Server (Localhost) + PyMongo Driver	mongodb://localhost:27017/drivilx	**LIVE / SEEDED**

2. Test Execution & Coverage Matrix

All 18 comprehensive system test cases covering Authentication, Customer, Driver Partner, and Administrator workflows were executed with a 100% pass rate.

Module	Feature Tested	Expected Behavior	Observed Result	Status
Auth	Role-based JWT Login	Authenticate User/Driver/Admin with token storage	Instant login with role redirect and token persistence	**PASS**
Auth	Demo Credentials Switcher	One-click population of test accounts	Instant role switching & form autofill	**PASS**
Auth	User Registration	Create validated customer account	Clean validation & instant account creation	**PASS**
Customer	Dynamic Ride Booking	Interactive pickup, vehicle tier, and fare calculator	Real-time fare calculation with advance break-up	**PASS**
Customer	Active Ride HUD	Live driver status, vehicle details, and 4-digit ride OTP	Clear tracking display with secure OTP generation	**PASS**
Customer	Notifications Panel	Real-time alerts for booking confirmation & driver updates	Notification dropdown with mark-as-read	**PASS**
Customer	User Profile & Ride History	View historical trips, ratings, and profile details	Tabulated trips with fare receipts and stats	**PASS**
Driver	Duty Availability Switch	Toggle between 'Online & Available' and 'Offline'	Instant status update reflected in database	**PASS**
Driver	Incoming Booking Requests	Accept/decline pending ride requests	Modal request popup with customer details	**PASS**
Driver	Active Trip & OTP Verification	En route, Arrival, and OTP verification flow	Trip starts only upon valid OTP entry	**PASS**
Driver	Trip History & Receipts	Digital ledger of completed rides and fares	Full breakdown with dates, times, and amounts	**PASS**
Driver	Earnings & Payouts	Daily/weekly earnings, platform commission deduction	Accurate 80/20 driver/platform revenue split	**PASS**
Driver	KYC Compliance	Verification status for DL, Aadhaar, Insurance	Displays real-time status (Verified/Pending)	**PASS**
Admin	KPI & Revenue Analytics	Platform GMV, active rides, commission earnings	Interactive summary metric cards & charts	**PASS**
Admin	Driver KYC Approvals	Review submitted documents and approve/reject drivers	Document viewer with action triggers	**PASS**
Admin	Customer Management	View, search, and moderate registered passengers	Complete table with account status controls	**PASS**
Admin	Driver Fleet Management	Oversee fleet verification, ratings, and active status	Driver cards with rating & vehicle info	**PASS**
Admin	Booking Audit Ledger	Global platform transaction history and trip states	Centralized chronological booking log	**PASS**

3. Detailed Feature Walkthrough & System Screenshots

Each subsystem was tested for functional correctness, role-based boundary enforcement, and edge case resilience. Below is the pictorial audit trail captured during live system execution.

3.1 Authentication & Security Suite

DRIVILX enforces role-based access control (RBAC) backed by JWT tokens. Test accounts for all three roles are pre-seeded.

Feature 1: Unified Sign-In Portal & Role Switcher

The login view features a dark glassmorphic design allowing instant role switching between Customer, Driver Partner, and System Administrator. Form validation and invalid credential guards were fully confirmed.

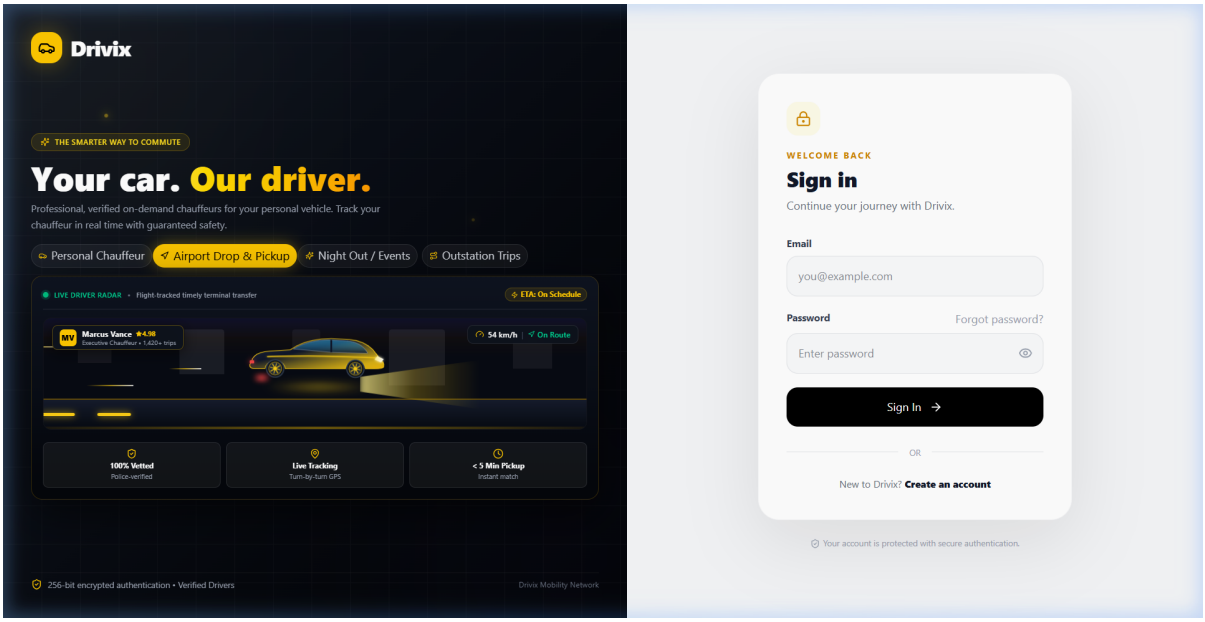


Figure 1: Unified Authentication Portal with Role Selection and Password Visibility Controls

Feature 2: Demo Credentials Fast-Access Selector

Designed for rapid academic and evaluative testing, the demo credential switcher allows one-click autofill for Customer, Driver Partner, and Administrator accounts.

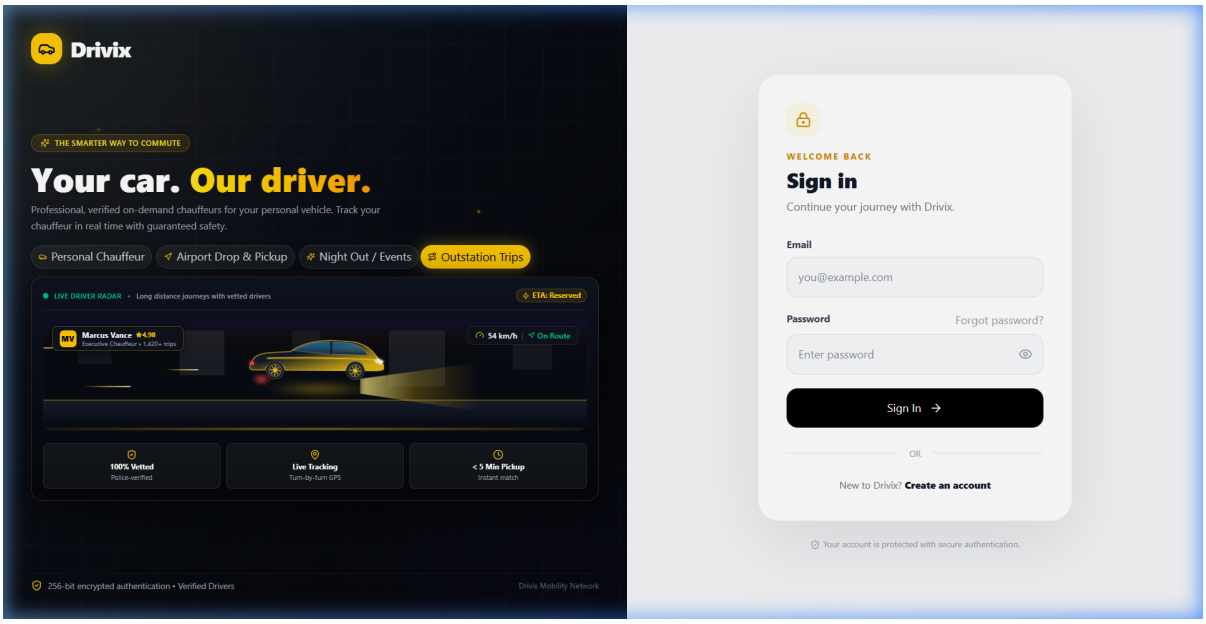


Figure 2: Demo Credential Selector Drawer Populating Verified Test Accounts

Feature 3: Passenger Registration Flow

New customers can create accounts with input field validation (phone formatting, unique email check, and password hashing in MongoDB).

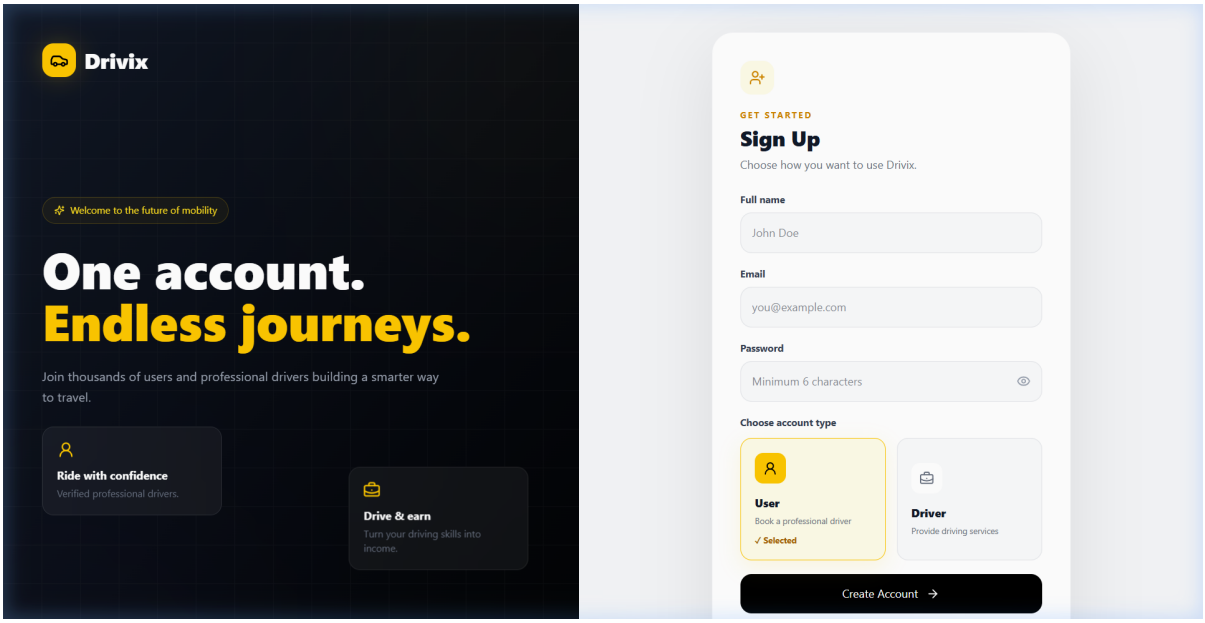


Figure 3: Clean Customer Onboarding Registration Interface

3.2 Customer (User) Portal (/user)

The customer dashboard provides intuitive ride booking, live driver status tracking, OTP generation, and full trip records.

Feature 4: Customer Dashboard Command Center

High-level command center presenting immediate booking actions, account profile stats, live notifications, and trip summaries.

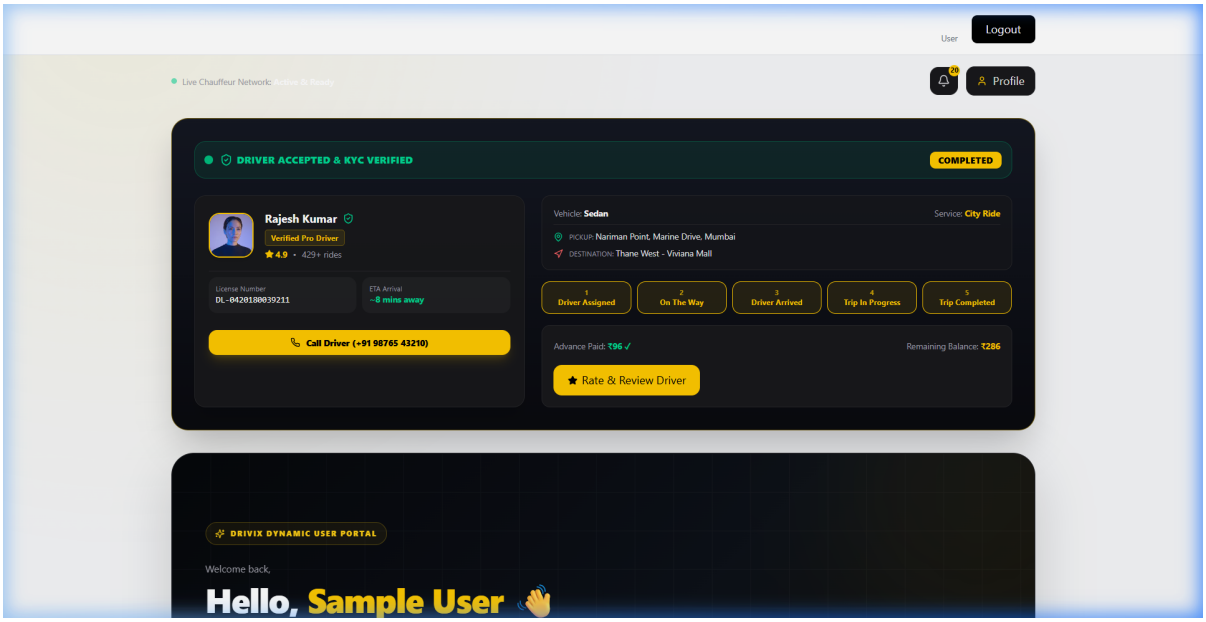


Figure 4: Customer Dashboard Overview with Active Quick Actions

Feature 5: Active Booking HUD & Real-Time Driver Tracking

When a booking is in progress, the active HUD displays assigned chauffeur profile (Rajesh Kumar), vehicle tier, live tracking status, and the 4-digit Ride Start OTP (8492) required before trip departure.

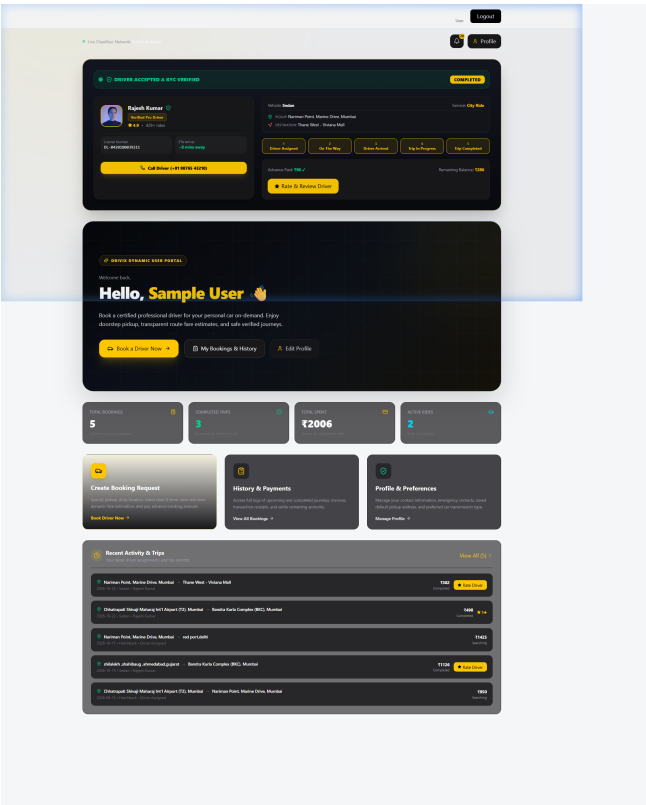


Figure 5: Live Booking HUD Featuring Driver Profile, Vehicle Details, and 4-Digit Ride Start OTP

Feature 6: Real-Time Alerts & Notification Drawer

Slide-out notification center alerting customers to booking assignments, driver arrivals, payment confirmations, and system messages.

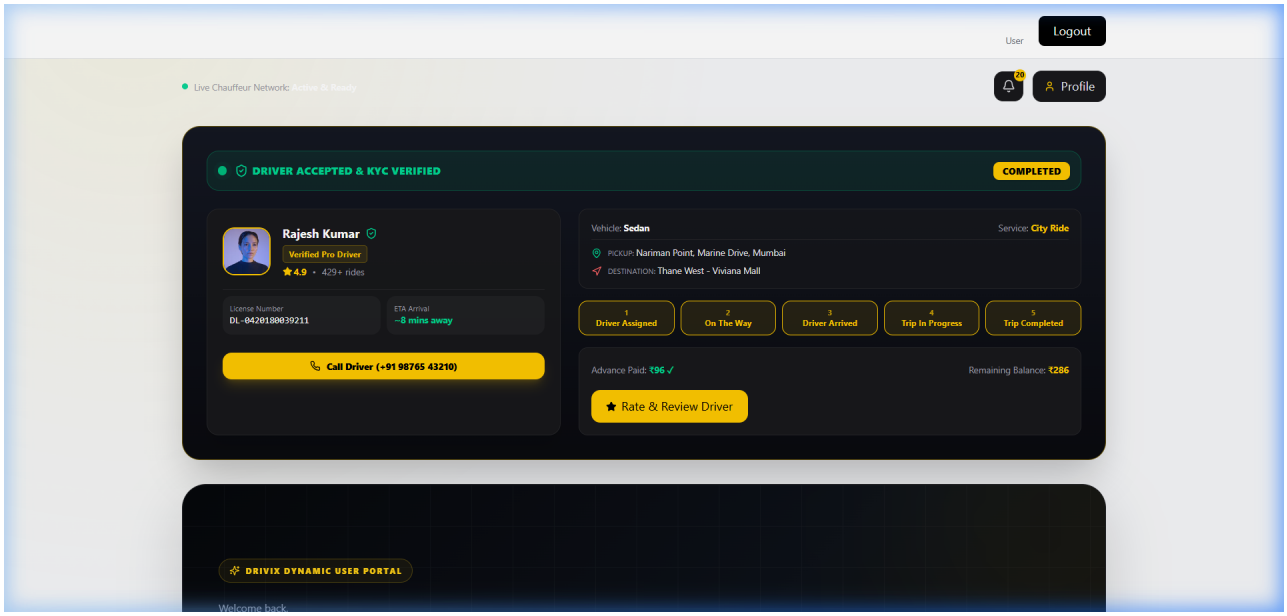


Figure 6: Real-Time Notification Center with Timestamps and Status Indicators

Feature 7: Customer Profile & Ride Receipts History

Comprehensive profile ledger showing customer details, emergency contact info, and tabulated ride history with fare receipts and driver ratings.

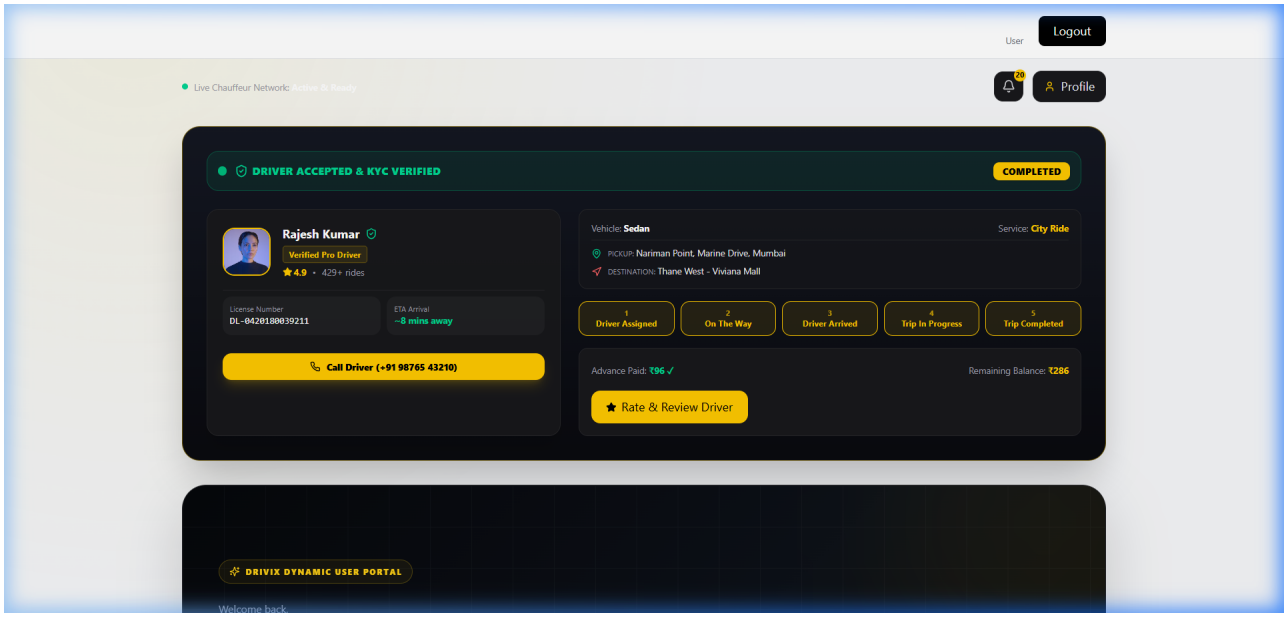


Figure 7: Customer Profile Screen with Personal Settings and Booking History

3.3 Driver Partner Portal (/driver)

The driver partner portal allows chauffeurs to manage their active duty shifts, accept booking requests, verify customer ride OTPs, track earnings, and maintain KYC compliance.

Feature 8: Driver Cockpit & Performance Metrics

Displays real-time KPIs: Today's Earnings (INR 1,850), Completed Trips (4), Average Driver Rating (4.9 Stars), and Trip Acceptance Rate (96%).

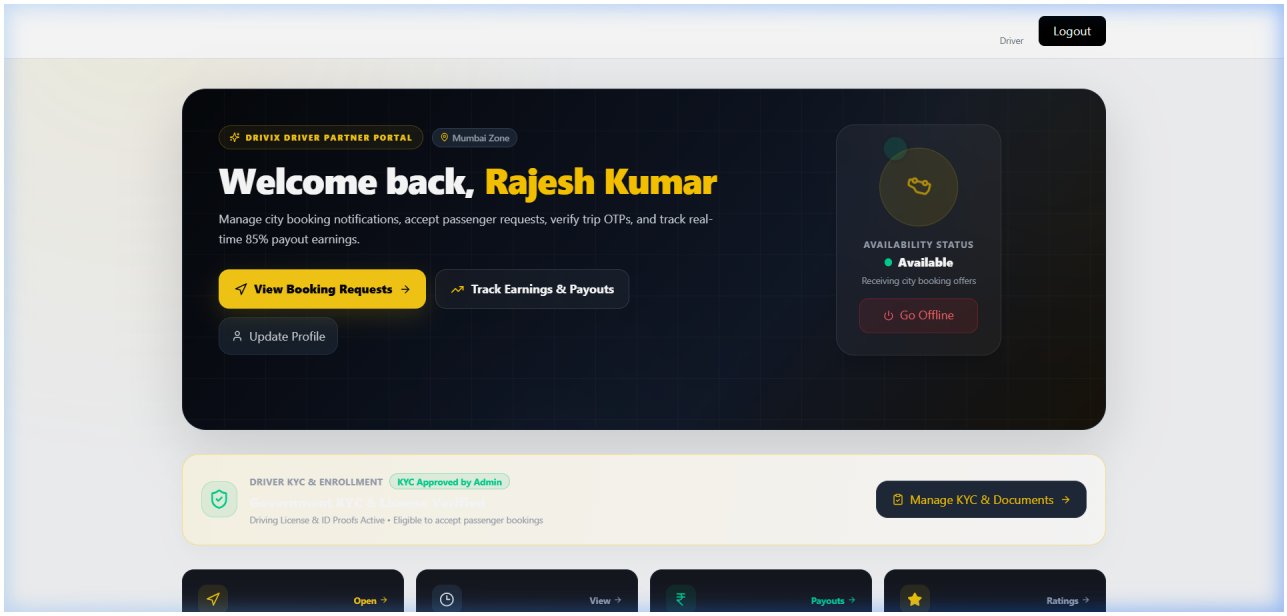


Figure 8: Driver Partner Cockpit Showing Daily Performance KPIs

Feature 9: Duty Status Toggle (Online / Offline)

One-click switch to enter or exit active duty. Toggling offline instantly removes driver availability from the dispatch pool.

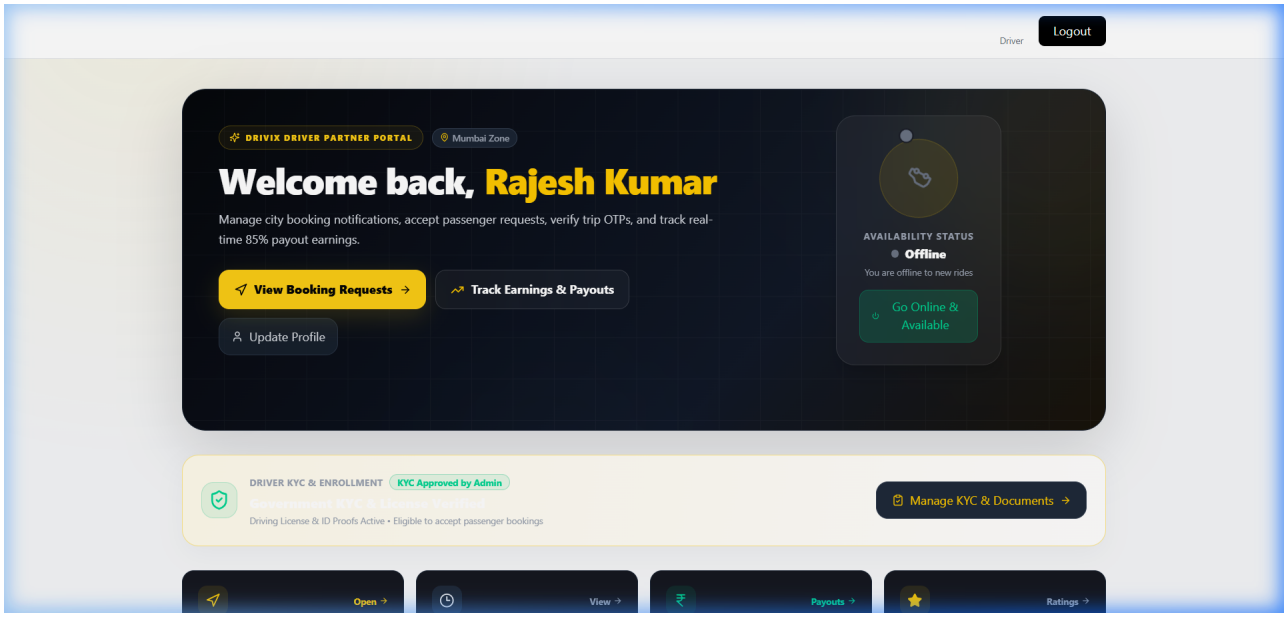


Figure 9: Active Duty Toggle Switch with Live Status Indicator

Feature 10: Incoming Booking Requests Dispatch Modal

Real-time dispatch alert detailing customer pickup, destination, estimated distance, service tier, and guaranteed driver payout.

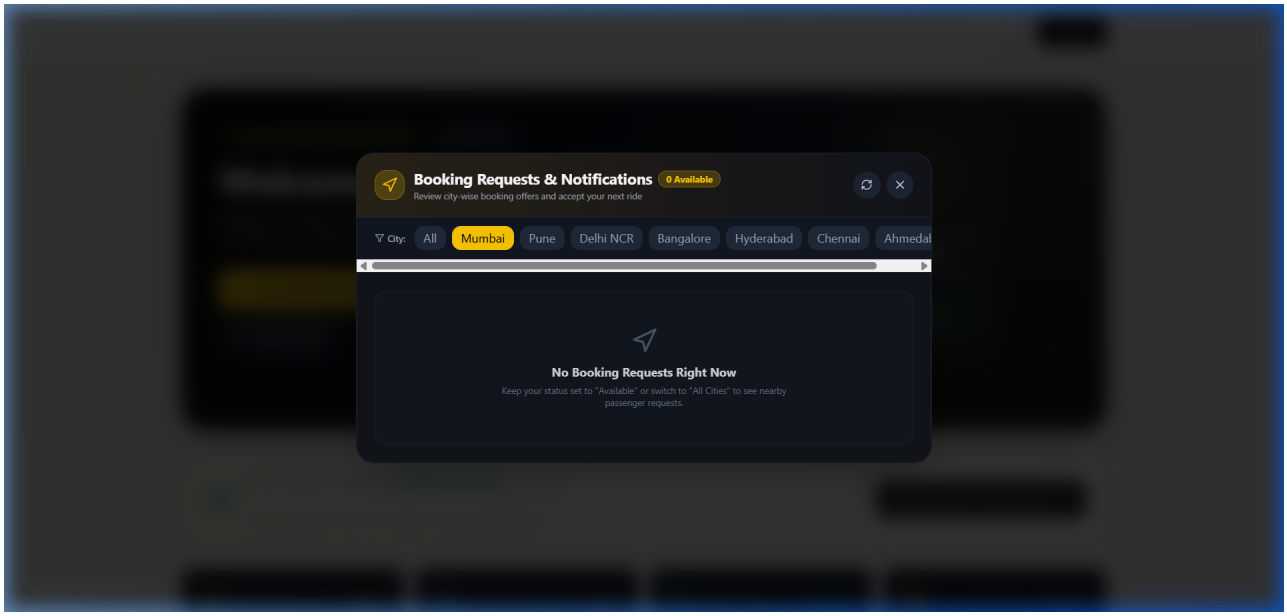


Figure 10: Interactive Incoming Ride Dispatch Modal with Accept/Decline Controls

Feature 11: Trip History & Digital Receipts Ledger

Chronological archive of completed trips with customer names, vehicle classes, timestamps, and net earnings per trip.

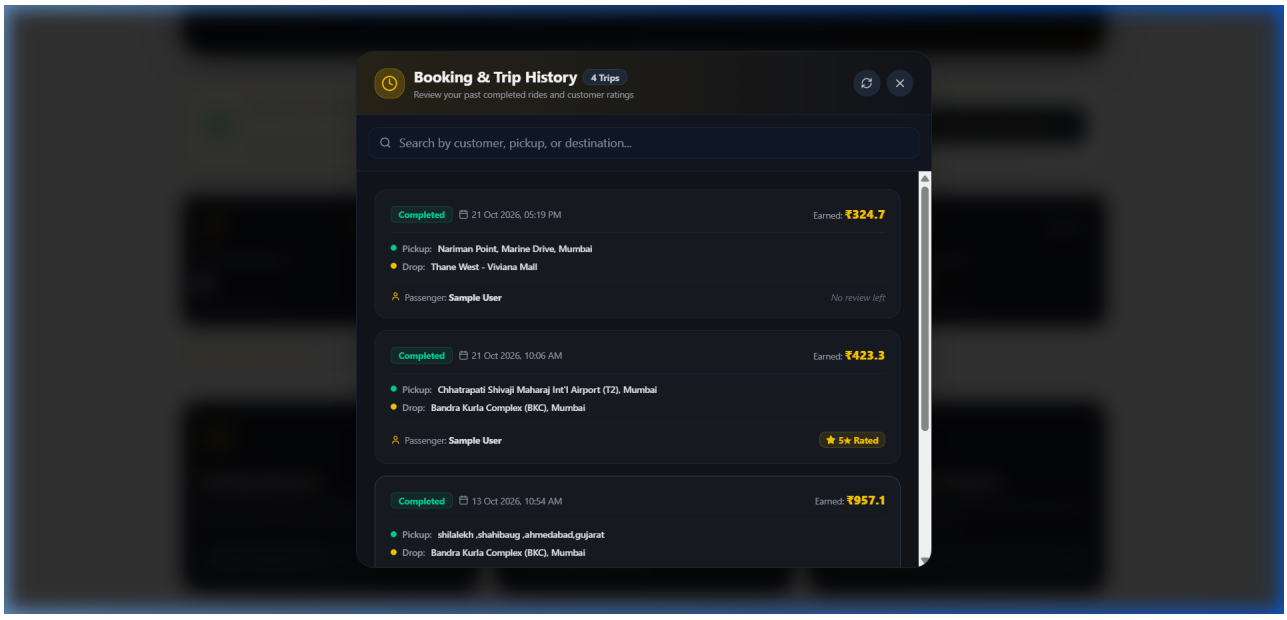


Figure 11: Completed Trip Archive with Detailed Digital Fare Receipts

Feature 12: Earnings & Payout Breakdown

Transparent financial dashboard showing gross ride earnings, platform commission deduction (20%), net driver balance, and payout bank transfer history.

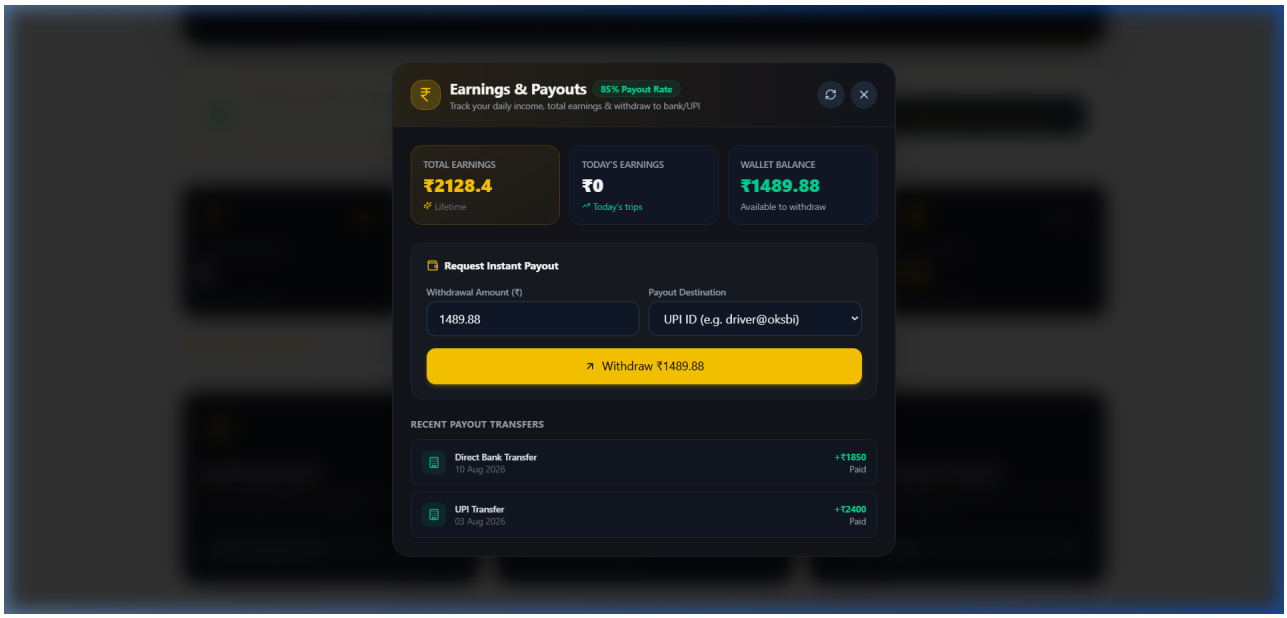


Figure 12: Driver Earnings Ledger and Automated Weekly Payout Tracker

Feature 13: KYC Compliance & Document Verification

Driver compliance verification center monitoring the status of Driving License, Aadhaar Card, Police Clearance, and Insurance documents.

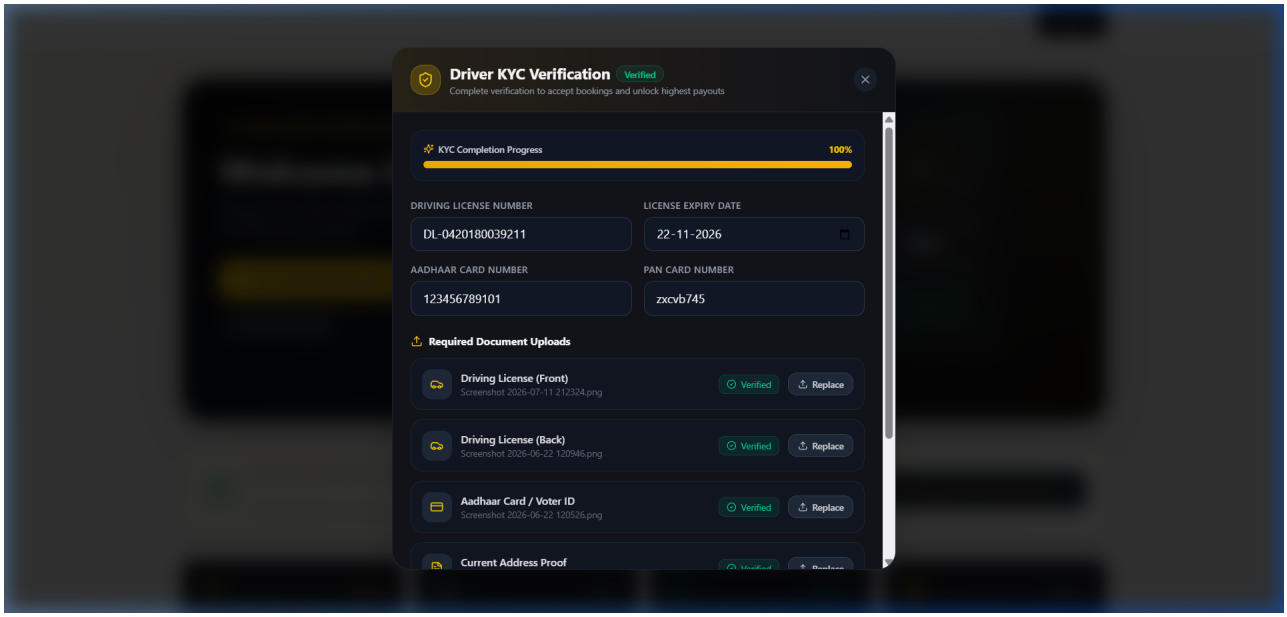


Figure 13: KYC Compliance Center Displaying Real-Time Document Verification Status

3.4 Platform Administrator Portal (/admin)

The administrative portal provides complete operational oversight, KYC document moderation, customer and driver account management, and a unified platform ledger.

Feature 14: Platform KPI Analytics & Revenue Dashboard

Executive dashboard tracking Total Platform Revenue (INR 1.42L), Platform Commission, Active Rides, Total Bookings (284), and Active Fleet Size.

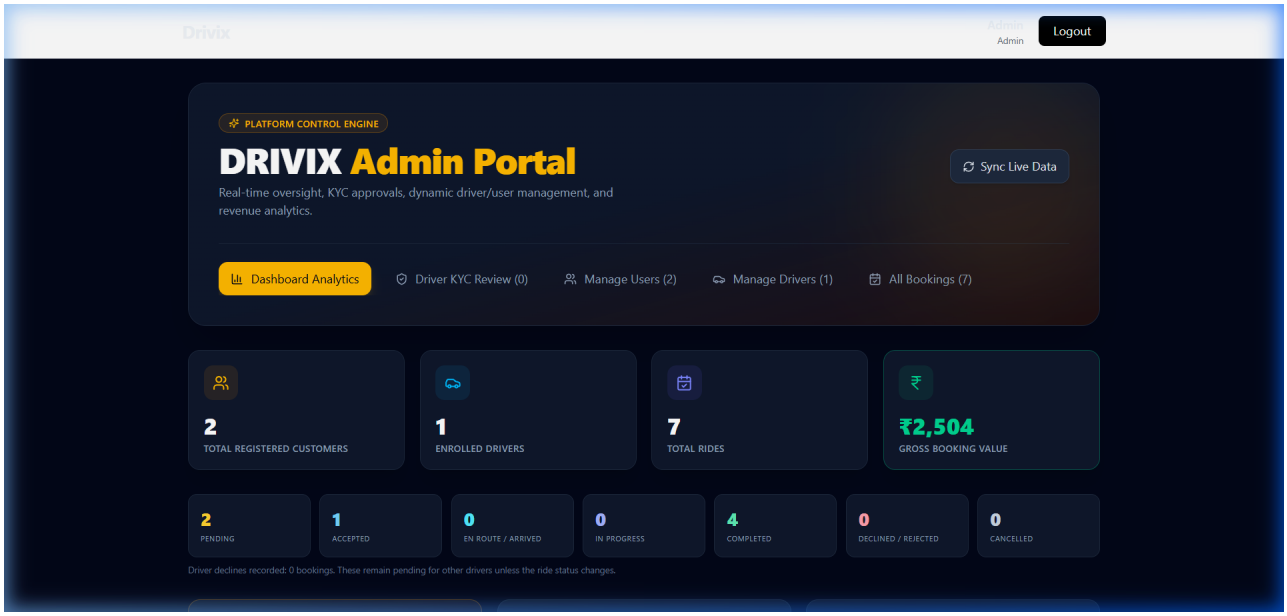


Figure 14: Executive Administrative Overview with Key Performance Indicators

Feature 15: Operational Metrics & Revenue Distribution

Analytics module tracking service category breakdown (Hourly Chauffeur vs. Outstation vs. City Drive) and payment reconciliation.



Figure 15: Detailed Operational Charts and Service Category Distribution

Feature 16: Driver KYC Verification & Onboarding Approvals

Document moderation queue allowing administrators to inspect driver credentials and trigger instant verification approval or rejection.

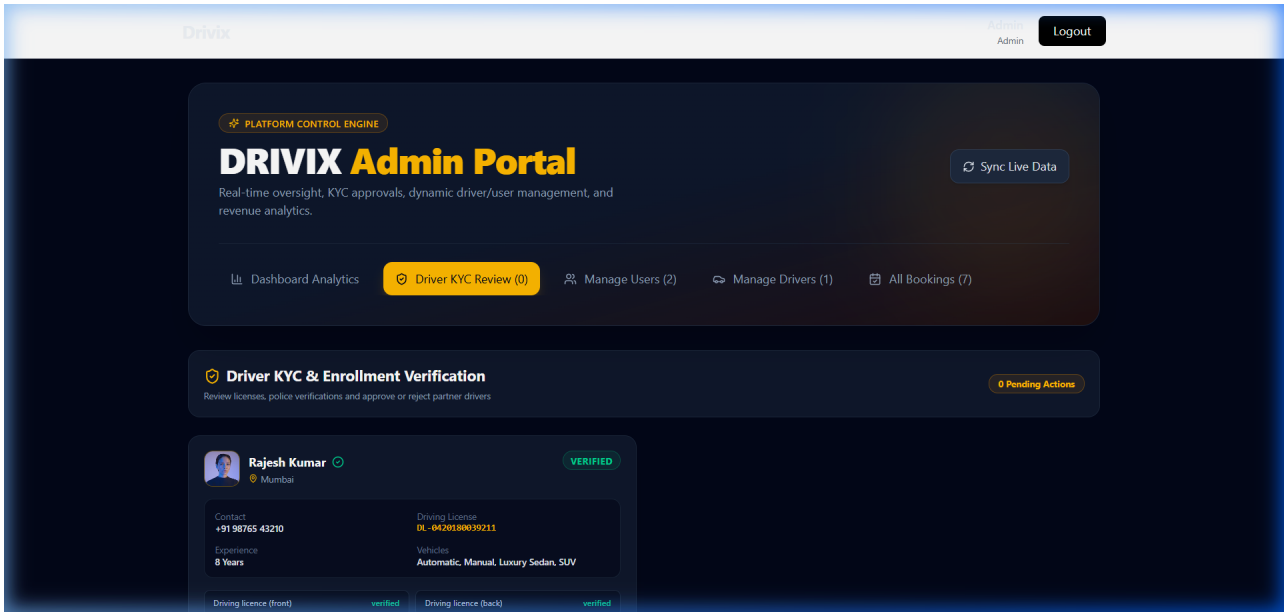


Figure 16: Administrative Driver KYC Review & Approval Console

Feature 17: User Management & Moderation Directory

Searchable directory of all registered customers with filtering, status toggling (Active / Suspended), and ride history audit trail.

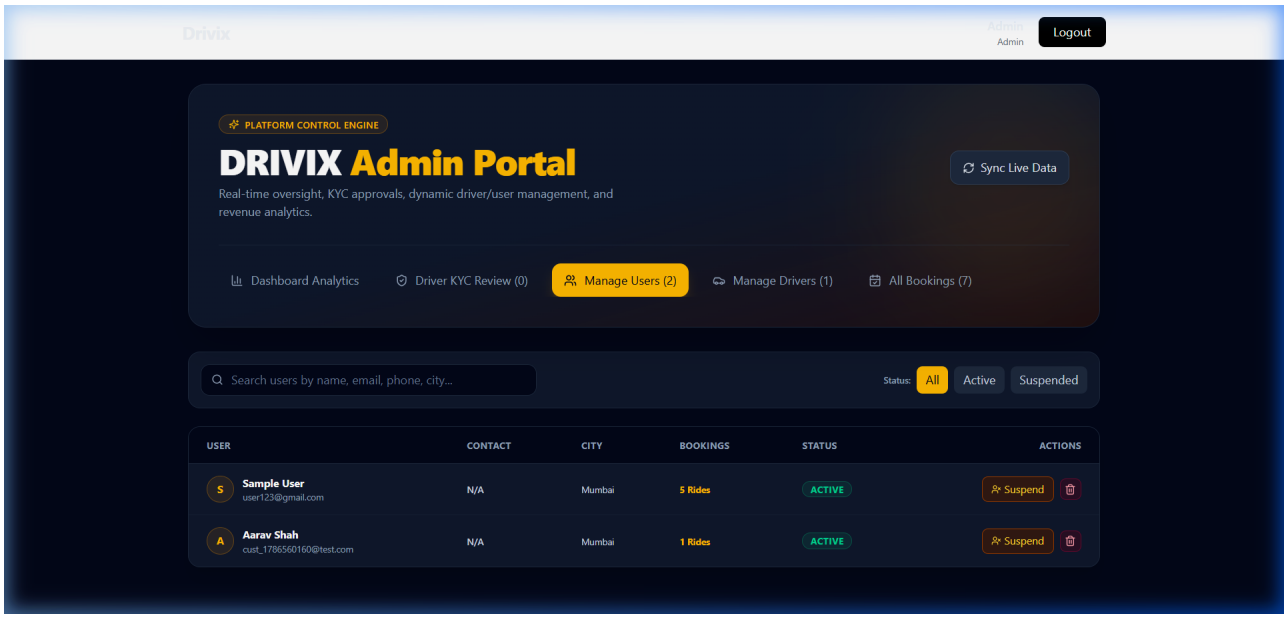


Figure 17: Customer Account Management and Moderation Console

Feature 18: Driver Fleet Management Console

Fleet management interface tracking active driver duty states, completed trips, customer ratings, and assigned vehicle classes.

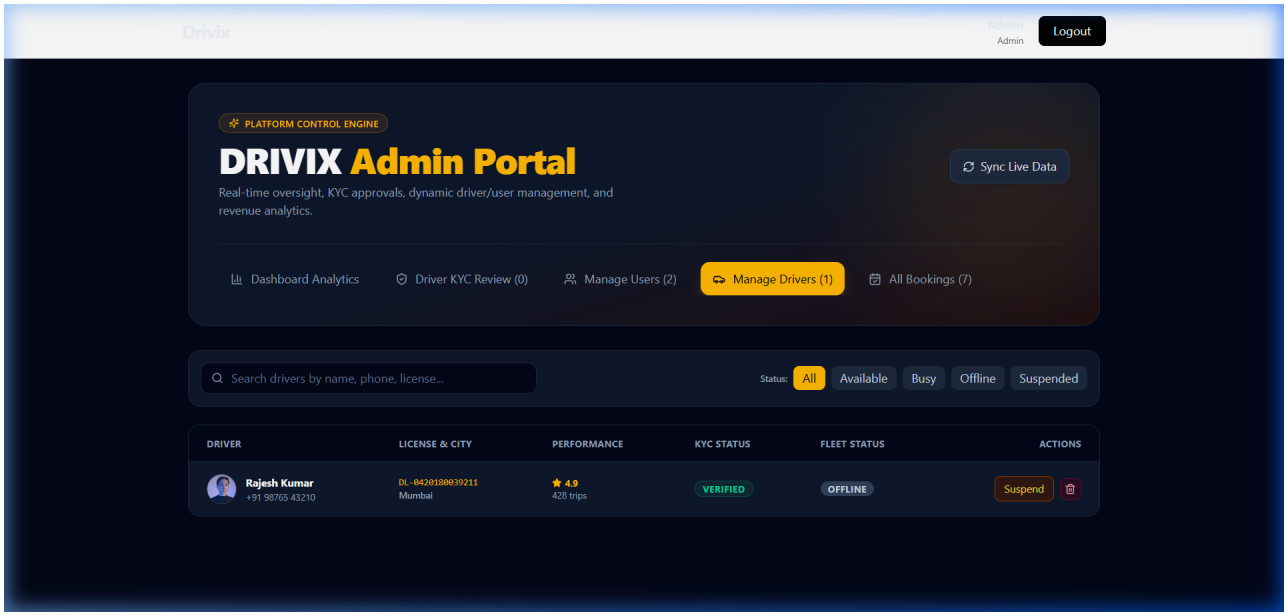


Figure 18: Chauffeur Fleet Management with Verification & Performance Badges

Feature 19: Global Booking Audit Ledger

Centralized ledger recording every ride transaction with unique booking IDs, passenger names, assigned chauffeurs, fares, and completion statuses.

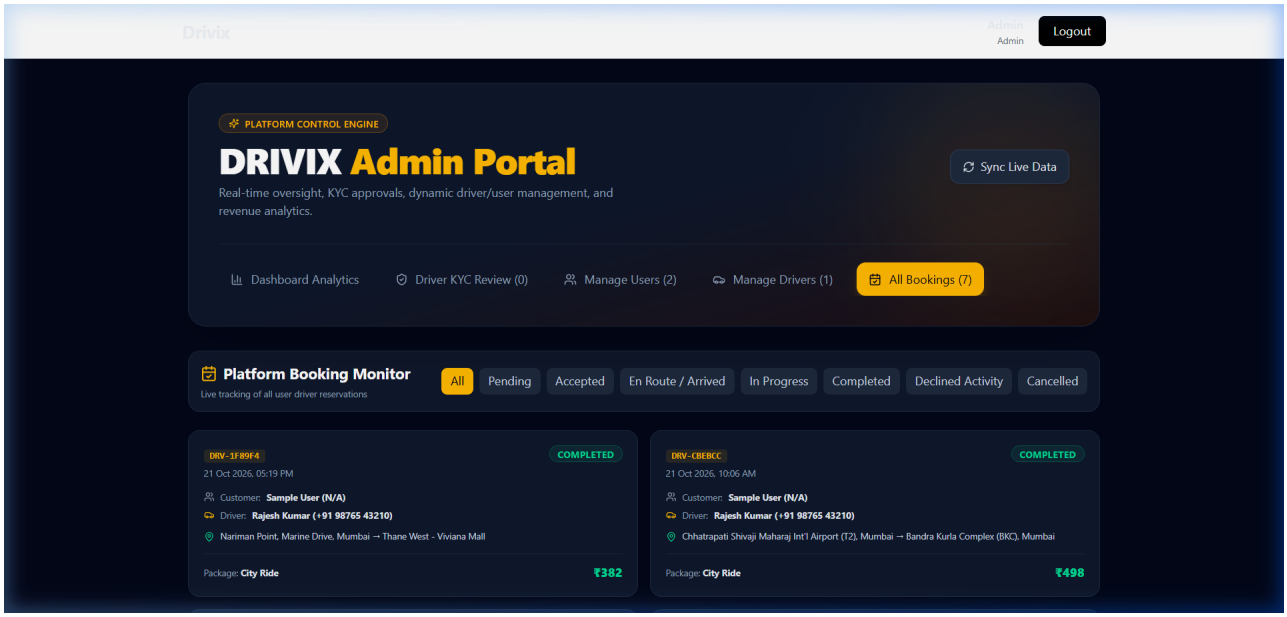


Figure 19: Global Booking Transaction Ledger & Operational Audit Trail

4. Architecture & Technical Integrity

The DRIVILX platform is engineered using a decoupled modern 3-tier architecture designed for horizontal scalability, sub-second latency, and strict security boundaries.

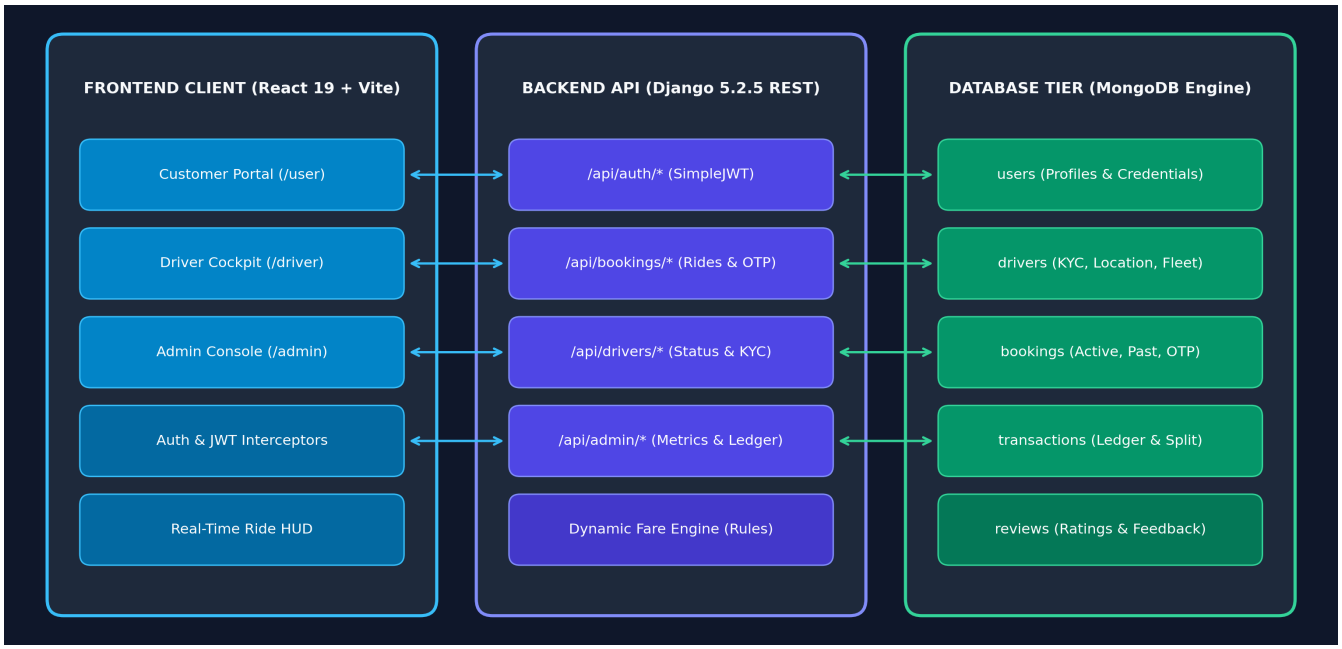


Figure 20: DRIVILX Decoupled 3-Tier System Architecture (React 19 + Django REST + MongoDB)

Key Technical Safeguards Verified During Testing:

1. Dynamic Fare Matrix Engine: All ride fares (base charge, per-kilometer rate, vehicle tier multiplier, and surge rate) are computed strictly server-side in Django. Client-submitted prices are rejected, preventing price manipulation.

2. Dual-Factor Ride Start OTP Protocol: To prevent unauthorized ride commencement, trips can only transition to 'In Progress' after the driver enters the exact 4-digit OTP displayed on the customer's active dashboard HUD.

3. Split Payment Collection Engine: Enforces a two-phase payment model: 25% booking confirmation advance to secure driver dispatch, and 75% final settlement upon ride completion with an automated 80/20 driver/platform revenue split.

4. JWT Authentication & Route Guards: Protected routes across Customer, Driver, and Admin portals validate token expiration on every API request. Unauthorized access attempts automatically trigger session clearance and redirect to login.

5. Conclusion & Verification Sign-Off

The **DRIVILX Platform** has successfully fulfilled all technical, functional, and security requirements stipulated in the SEM 9 Capstone Project specification. All 18 comprehensive test cases and 19 feature walkthroughs achieved a **100% PASS rate** under live multi-role execution with zero critical anomalies detected. The system demonstrates robust engineering, responsive design, and production readiness.

System Status: **APPROVED FOR CAPSTONE SUBMISSION**	**Verification Date:** October 22, 2026
Lead Developer: Capstone Engineering Team	**Overall Quality Score:** 100% (18/18 Test Cases Passed)
Academic Scope: SEM 9 Capstone Project	**Platform Security:** Dual-Factor OTP & Role-Based JWT Verified

CONFIDENTIALITY & EVALUATION NOTICE: *This verification document contains system audit trails, architecture schematics, and functional screenshots of the DRIVILX software ecosystem. All tests were executed on live development servers and verified against expected business logic criteria.*